

		Pd across R = $50/20 = 2.50$ V
23	A	The forces acting on XY and ZW have the same magnitude in opposite directions. These two forces form a couple with a torque as follow: $\tau = NBIL \times d = (20)(0.80)(I)(0.17) \times (0.11) = 1.35$ $I = 4.5 \text{ A}$
24	A	Negative charges gain KE as it loses EPE from Source S to the hollow metal